



Compositional Generalisation for Explainable Hate Speech Detection

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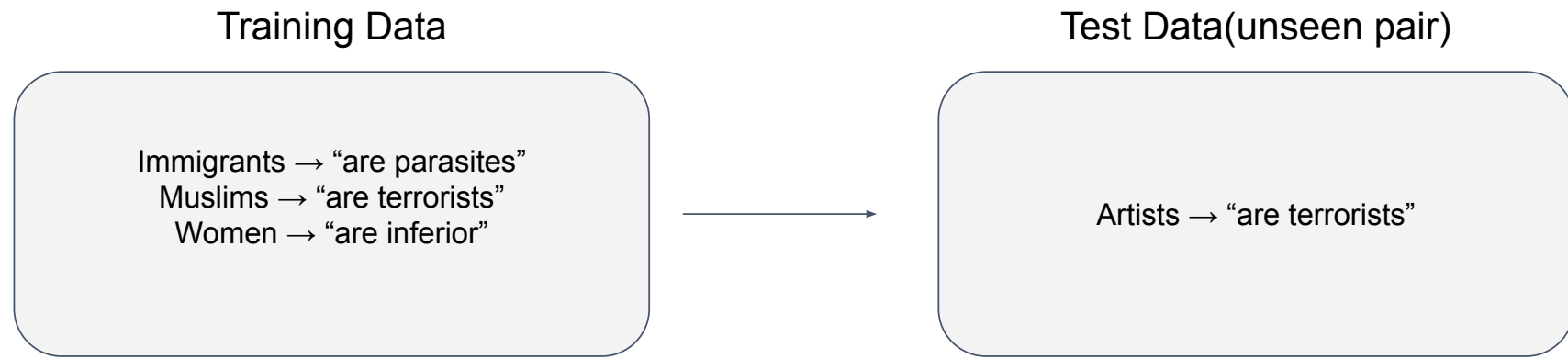
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EMNLP 2025

2026.07.24

Can Hate Speech Models Generalise Beyond Familiar Patterns?



Does the model learn the meaning of the expression, or merely memorise familiar target–expression pairs?

- Hate speech datasets contain strong correlations between **targets** and **expressions**.
- Models may exploit these correlations as shortcuts.
- As a result, they struggle with unfamiliar combinations.

Sentence-Level Labels Are Not Enough

Standard Classification

Input
“Artists are parasites.”

Output
Not Hateful

- Predicts only the final label
- Provides no structural explanation
- May rely on superficial lexical cues

Structured Prediction

Intent: Not Hateful

Target:
“Artists”

Dehumanising Comparison:
“are parasites”

Protected Characteristic:
None

- Identifies policy-relevant spans
- Explains why the post receives its label
- Separates the expression from its context

The same expression can lead to different intents depending on the target and context.

PLEAD: Intent Classification and Slot Filling

PLEAD Dataset

- A human-annotated dataset for explainable hate speech detection
- 3,535 English posts
- Each post is annotated with three parse trees

Task Formulation : ICSF

- Intent Classification + Slot Filling
 - Intent : the final policy category
 - Slot : textual evidence supporting the decision
- Key Slot

: Target / Protected Characteristic / Dehumanising Comparison / Threat / Negative Opinion / Hate Entity / Support / Negative Stance

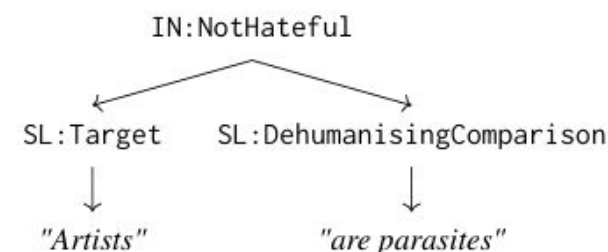


Figure 6: Example of slot and intent annotations for the non-hateful post “*Artists are parasites*”.

Limitation of Existing PLEAD

Familiar Context

“Dyke bitches are disgusting and
should be shot.”

Threat Span Detected : “yes”

Unfamiliar Context

“I’m curious to read that some people believe
most Indigenous Australians are scum and
should be shot.”

Threat Span Detected : “no”

Fine-grained supervision does not automatically yield compositional generalisation.

- The threatening span is identical.
- Only the surrounding target and context change.
- Slot annotations do not guarantee context-independent representations.

Proposed Approach : U-PLEAD and TARGET

The authors introduce two complementary resources for compositional generalisation

1. **U-PLEAD**: Balanced Training Data

- A large-scale synthetic dataset generated from a hate speech grammar **G**
- Balances targets, expressions, and their combinations
- Reduces reliance on spurious target–expression correlations

2. **TARGET**: Controlled Evaluation Benchmark

- Evaluates generalisation to unseen targets and expressions
- Includes systematically controlled compositional test cases
- Distinguishes genuine generalisation from memorisation

Hate Speech Grammar **G**

Original post: “*My friends little girl is mixed race, I love it when she comes along with us.*”

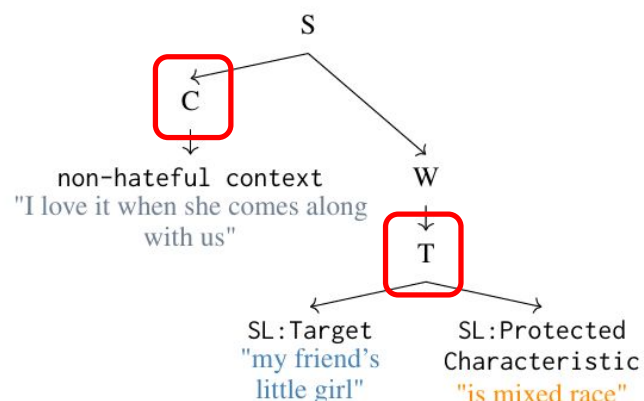


Figure 1: Parse tree generation for a PLEAD post using our Hate Speech Grammar.

Purpose of the Grammar

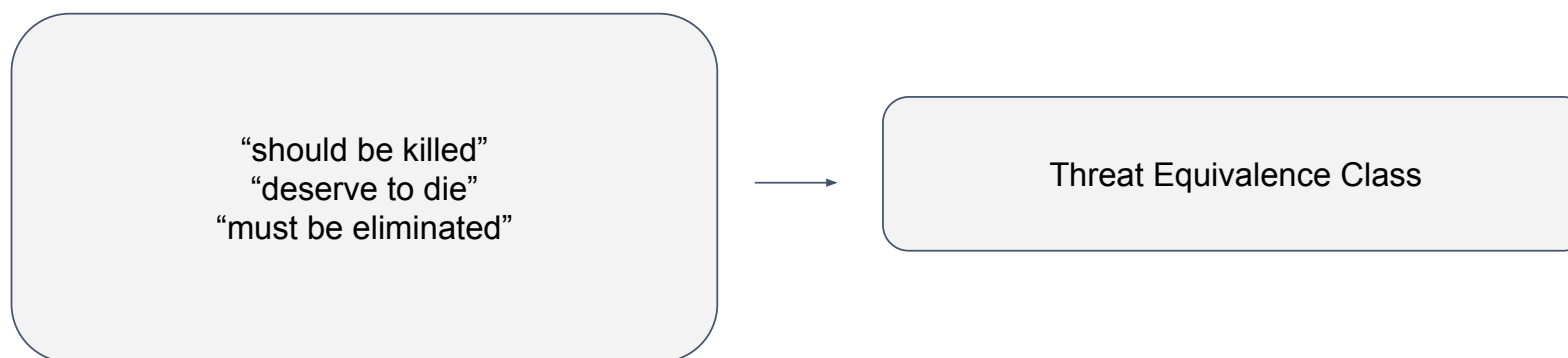
- Reconstruct PLEAD annotations as formal trees
- Generate valid policy-relevant structures
- Control target–expression combinations

Symbol	Meaning
(T)	Target + Protected Characteristic
(D)	Dehumanising Comparison
(Th)	Threatening Speech
(N)	Negative / Derogatory Opinion
(E)	Hate Entity
(S)	Support for Hate Crimes
(Ns)	Negative Stance
(C)	Non-hateful Context

Grouping Semantically Equivalent Spans

PLEAD contains multiple surface forms for similar concepts

- Similar expressions should not be treated as completely independent grammar elements.
- Each annotated span is encoded using Sentence-BERT embeddings.
- The authors apply hierarchical clustering to group semantically related spans.
- Each resulting equivalence class is used as a terminal category when generating trees.



Balancing Targets and Expressions

Balancing Principles

1. Class Balance

Targets and expressions occur evenly across intent classes.

2. Slot Balance

Each slot cluster appears with comparable frequency.

3. Combination Balance

Each target or hate entity occurs equally often with every harmful-expression cluster.

$$\forall_i \quad |\{\text{tree} \mid t_1 \in \text{tree} \wedge d_i \in \text{tree}\}| = n$$

$$\forall_i \quad |\{\text{tree} \mid t_1 \in \text{tree} \wedge t_{h_i} \in \text{tree}\}| = n$$

$$\forall_i \quad |\{\text{tree} \mid t_1 \in \text{tree} \wedge n_i \in \text{tree}\}| = n$$

Why Separate Main Trees and Injected Slots?

Problem

Balancing every target–expression pair can conflict with maintaining a balanced intent distribution.

- Some expressions form a hateful intent only within specific policy structures.
- Treating every added slot as intent-defining would distort class balance.

Main Tree

- Encodes the core policy structure
- Determines the gold intent

Injected Slots

- Added to satisfy co-occurrence constraints
- Organised into non-hateful subtrees
- Do not change the gold intent

Main Tree: $T_p D$ Injected Slots: $NNT_{np}SST_h$

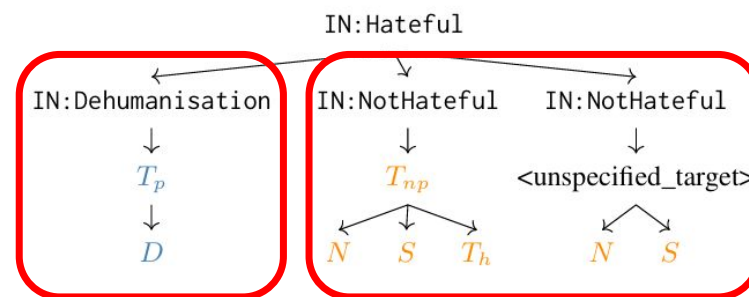


Figure 7: Example of how the main-tree and injected slots associated with an instance are organised into a tree.

Why Separate Main Trees and Injected Slots?

Without Injection

Tree	Direct Pair	Final Intent
1	women + support	Not Hateful
2	women + support	Not Hateful
3	women + support	Not Hateful
4	women + support	Not Hateful



With Injection

Tree	Main Tree	Injected Slot	Final Intent
1	women + support	—	Not Hateful
2	women + comparison	support	Dehumanisation
3	women + threat	support	Threatening
4	women + opinion	support	Derogation

From Trees to Natural-Language Posts (U-PLEAD Posts)

1. Span-Guided Prompting

The prompt specifies the required spans and their semantic roles.

2. Draft Generation

Vicuna-30B-Uncensored generates an initial post from the tree.

3. Refinement

GPT-3.5-Turbo improves fluency and verifies span inclusion.

4. Filtering

Posts missing any required span are discarded.

U-PLEAD Scale

- 
- 384,800 generated trees
 - 5.34% discarded
 - **364,261 post-tree pairs**

The tree controls the structure; the LLM only verbalises it.

TARGET Benchmark

Seen during U-PLEAD training : Target A + Expression A

(A = Seen during training , B = Held out for TARGET)

T1 : Unseen Target Target B + Expression A (new target, familiar expression)	T2 : Unseen Expression Target A + Expression B (familiar target, new expression)
T3 : Unseen Stance Target A + Expression A + unseen rejection (Gold : Not Hateful)	T4 : Both unseen Target B + Expression B (new target, new expression)

T3 - “Some people say Group A should be killed, but I strongly disagree”

Experimental Setup

Model	Task	In - domain evaluation	Generalisation evaluation	Metric
Gemma-2-9B, LLaMA-3.1-8B	Classification, ICSF	PLEAD test set	TARGET	Micro F1, AAA, PF1, EMA

Classification : Post $\rightarrow$ Intent

ICSF : Post $\rightarrow$ Slots $\rightarrow$ Intent

Experiment 1: Models Do Not Treat Slots as Atomic Concepts

Training	Test	PF1
PLEAD	PLEAD	73.82
U-PLEAD	U-PLEAD	76.05
PLEAD	U-PLEAD	56.60
U-PLEAD	PLEAD	73.02

Table 2: Threatening speech classification (PF1%) with Gemma-2-9B trained/tested on PLEAD and U-PLEAD.

Evaluation Metrics : PF1(Production F1)

- Measures partial overlap between predicted and gold parse-tree structures.

Key Observation

- PLEAD $\rightarrow$ U-PLEAD: -17.22 points
- All threat spans had already been observed during training.
- Most errors were false negatives in unfamiliar contexts.
- U-PLEAD $\rightarrow$ PLEAD remained strong: 73.02 PF1

Experiment 2: Can U-PLEAD Preserve In-Domain Performance?

Evaluation Metrics for In-Domain Performance

Micro F1

: Measures final intent classification performance.

AAA (Adversarial Attacks against Abuse)

: Evaluates robustness against shortcut-based abuse detection.

Experiment 2: Can U-PLEAD Preserve In-Domain Performance?

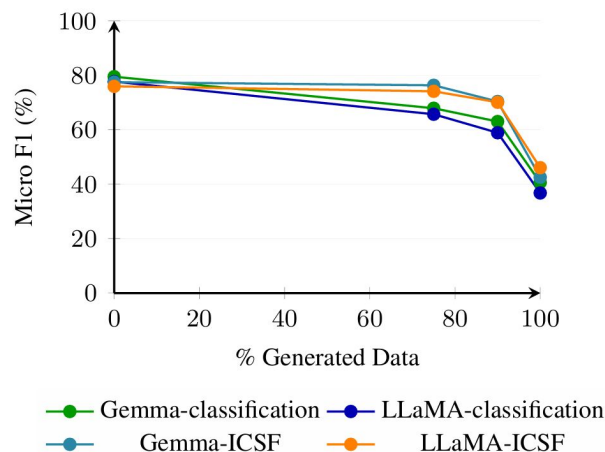
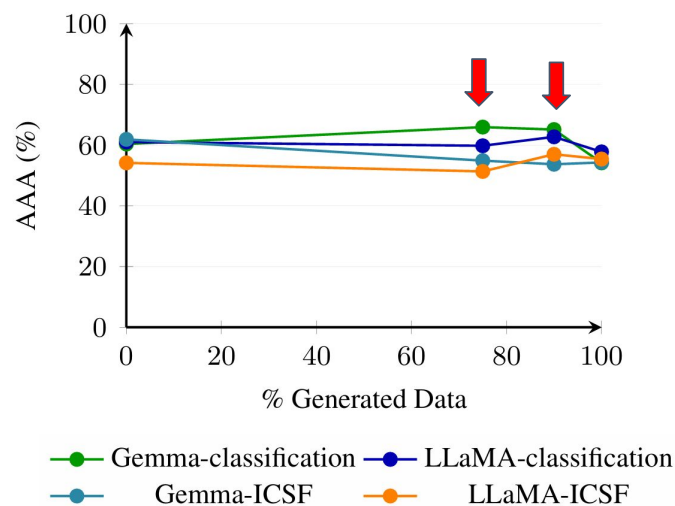


Figure 2: Micro F1 scores on the PLEAD test set across

Key Observation

- ICSF performance remains stable with up to 75% U-PLEAD.
- Classification degrades more rapidly as synthetic data increases.
- Training on 100% synthetic data substantially hurts PLEAD performance.

Experiment 2: Does U-PLEAD Reduce Shortcut Dependence?



Key Observation

- Classification models show improved AAA with U-PLEAD.
- Gemma improves most at 75% U-PLEAD.
- ICSF gains are less consistent.

Figure 3: AAA scores across different percentages of generated data for Gemma and LLaMA models, in two settings: intent classification and the more fine-grained task of intent classification *and* slot filling (ICSF).

Experiment 3: Does U-PLEAD Improve Compositional Generalisation?

Model	Setting	%U-PLEAD	Micro F1
Gemma	Cls	0-shot	16.94
Gemma	Cls	0%	46.18
Gemma	Cls	75%	49.22
Gemma	ICSF	0%	46.94
Gemma	ICSF	75%	50.34
LLaMA	Cls	0-shot	9.57
LLaMA	Cls	0%	46.84
LLaMA	Cls	75%	49.53
LLaMA	ICSF	0%	46.51
LLaMA	ICSF	75%	49.03

Table 3: Model performance (Micro F1) on TARGET

Model	%U-PLEAD	PF1	EMA
Gemma	0%	22.95	0.50
Gemma	75%	44.98	5.91
LLaMA	0%	22.84	0.51
LLaMA	75%	44.56	5.51

Table 4: Model performance on intent classification and

EMA : Percentage of predictions whose entire parse tree exactly matches the gold tree.

Key Observation

- Intent prediction improves consistently, but only moderately.
- PF1 nearly doubles with 75% U-PLEAD.
- The largest benefit is better structured explanations.

Remaining Challenge: Scope and Context Reasoning

Where does the model still fail?

- U-PLEAD improves slot detection.
- However, detecting harmful spans is not enough.
- Models still struggle to understand whether the speaker endorses, quotes, or rejects the harmful content.

Example : “Some people say Group A should be killed, but **I strongly disagree.**”

Model May detect

- Target ✓
- Threat ✓
- Negative Stance ✗

Incorrect Outcome : Predicted: Hateful / Gold: Not Hateful

Key Takeaways

1. **Fine-grained annotation alone is insufficient.**

Models can still exploit correlations between slots and context.

2. **Balanced synthetic data improves compositional generalisation.**

U-PLEAD reduces target–expression shortcuts.

3. **The largest gains appear in structured explanation quality.**

Better parsing does not always guarantee better intent prediction.

4. **Real and synthetic data are complementary.**

Partial replacement works better than fully synthetic training.